# Lecture 7: Continuity

Mathematical Analysis

## 1 Continuity

**Definition 1.1** (Function Continuity at a Point)**.** [def:function-continuity] Under metric space $(A, d_A)$ and $(B, d_B)$, let $f : A \to B$. $f$ is continous at a point $c \in A$ if

$$\forall \varepsilon : \mathbb{R}_+, \exists \delta : \mathbb{R}_+ \, \forall x \in B_\delta(c), \, d_B(f(x), f(c)) < \varepsilon$$

**Lemma 1.1.** *If a function $f$'s limit exists at a point $c \in A$ which is $\ell \in B$, and $f$ is continuous at $c$ and $\ell = f(c)$*

**Lemma 1.2.** *Every definition isolated points[isolated point] $c$ of a set $A$ is*